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Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST 09** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 1: Real Numbers** | **MM: 30**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit theorem/formula declaration, complete prime factorisation or contradiction steps, and boxed final answers with units.
2. In Assertion–Reason questions, truth values of both (A) and (R) as well as whether (R) validly explains (A) are explicitly established.
3. Partial marks are allocated per step in brackets: **[0.5M, 1M, 2M]**. Alternative valid methods receive full credit.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: MCQs & Assertion–Reason

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Q1. If $a = 2^3 \times 3^2$ and $b = 2^2 \times 3^3$, then the value of $\dfrac{\text{LCM}(a,b)}{\text{HCF}(a,b)}$ is: [1]

(a) 6 (b) 12 (c) 36 (d) 18

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (a) 6

We know the prime-exponent rules for HCF and LCM:

Formula Used: $\text{HCF}(a,b) = 2^2 \times 3^2, \quad \text{LCM}(a,b) = 2^3 \times 3^3$

Dividing: $\dfrac{\text{LCM}(a,b)}{\text{HCF}(a,b)} = \dfrac{2^3 \times 3^3}{2^2 \times 3^2} = 2^{3-2} \times 3^{3-2} = 2 \times 3 = \boxed{6}$. [1M]

Q2. If $x = 2^3 \times 3 \times 5^2$ and $y = 2^2 \times 3^3$, then $\text{HCF}(x,y)$ is equal to: [1]

(a) 12 (b) 24 (c) 36 (d) 48

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (a) 12

Common prime factors of $x = 2^3 \times 3^1 \times 5^2$ and $y = 2^2 \times 3^3$ are 2 and 3. We know:

Formula Used: $\text{HCF}(x,y) = \prod p^{\min \text{ exp}}$

$\text{HCF}(x,y) = 2^{\min(3,2)} \times 3^{\min(1,3)} = 2^2 \times 3^1 = 4 \times 3 = \boxed{12}$. [1M]

Q3. The number of consecutive trailing zeros in the number $N = 2^3 \times 3^4 \times 5^4 \times 7$ is: [1]

(a) 3 (b) 4 (c) 7 (d) 12

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (a) 3

Each trailing zero is formed by a factor of $10 = 2 \times 5$. We know:

Formula Used: Trailing zeros in $2^a \times 5^b \times (\text{coprime to } 10) = \min(a, b)$

$N = (2^3 \times 5^3) \times 3^4 \times 5^1 \times 7 = 10^3 \times (3^4 \times 5 \times 7)$. Thus trailing zeros $= \min(3, 4) = \boxed{3}$. [1M]

Q4. The exponent of 2 in the prime factorisation of 144 is: [1]

(a) 2 (b) 3 (c) 4 (d) 6

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (c) 4

We factorise 144 into prime powers:

Formula Used: $144 = 16 \times 9 = 2^4 \times 3^2$

The exponent of prime 2 is $\boxed{4}$. [1M]

Q5. If p and q are positive integers such that $p = ab^3$ and $q = a^4b$, where a and b are prime numbers, then HCF(p, q) is: [1]

(a) ab (b) a^2b^2 (c) a^4b^3 (d) a^3b^2

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (a) ab

Given $p = a^1b^3$ and $q = a^4b^1$. We know:

Formula Used: HCF$(p, q) = a^{\min(1,4)} \cdot b^{\min(3,1)}$

Thus HCF$(p, q) = a^1b^1 = \boxed{ab}$. [1M]

Q6. **Assertion (A):** The HCF of two consecutive even natural numbers is 2.
Reason (R): Every even natural number is completely divisible by 2. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

Any two consecutive even natural numbers have the form $2k$ and $2(k + 1)$. Since HCF$(k, k + 1) = 1$, their HCF is $2 \cdot \text{HCF}(k, k + 1) = 2$, so (A) is true. Reason (R) is also true by definition of even integers. However, we know:

Formula Used: $2 \mid a$ and $2 \mid b$ only proves 2 is a common factor, not that HCF$(a, b) = 2$

(For instance, 4 and 8 are both even, yet HCF$(4, 8) = 4 \neq 2$.) Thus (R) alone does not explain why the HCF is exactly 2, giving option $\boxed{\text{(B)}}$. [1M]

Q7. Prove that $3+2\sqrt{5}$ is an irrational number, given that $\sqrt{5}$ is irrational. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: $\sqrt{5}$ is irrational. Suppose, on the contrary, that $3+2\sqrt{5}$ is rational, equal to $\frac{a}{b}$ $(a, b \in \mathbb{Z},\ b \neq 0)$. Isolating $\sqrt{5}$:

$$3+2\sqrt{5} = \frac{a}{b} \implies 2\sqrt{5} = \frac{a}{b} - 3 = \frac{a-3b}{b} \implies \sqrt{5} = \frac{a-3b}{2b}. \qquad [1M]$$

We know rational numbers are closed under subtraction and non-zero division:

Formula Used: $a, b \in \mathbb{Z},\ b \neq 0 \implies \frac{a-3b}{2b} \in \mathbb{Q}$

Hence $\sqrt{5}$ must be rational, which contradicts the given fact that $\sqrt{5}$ is irrational. Therefore $\boxed{3+2\sqrt{5} \text{ is irrational.}}$ [1M]

Q8. Prove that $5-\sqrt{3}$ is an irrational number, given that $\sqrt{3}$ is irrational. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: $\sqrt{3}$ is irrational. Suppose, on the contrary, that $5-\sqrt{3} = \frac{a}{b}$ is rational $(a, b \in \mathbb{Z},\ b \neq 0)$. Rearranging for $\sqrt{3}$:

$$5-\sqrt{3} = \frac{a}{b} \implies \sqrt{3} = 5 - \frac{a}{b} = \frac{5b-a}{b}. \qquad [1M]$$

We know the closure property of rationals:

Formula Used: $a, b \in \mathbb{Z},\ b \neq 0 \implies \frac{5b-a}{b} \in \mathbb{Q}$

This forces $\sqrt{3}$ to be rational, contradicting the given statement that $\sqrt{3}$ is irrational. Thus $\boxed{5-\sqrt{3} \text{ is irrational.}}$ [1M]

SECTION C: Short Answer Type MATHS MANTHAN | (2 × 3 = 6 Marks)

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Q9. In a school library, there are 780 books of English literature and 364 books of Hindi literature. The librarian wants to arrange them in shelves such that each shelf has the same number of books of the same language. Find the maximum number of books per shelf and the total number of shelves required. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Maximum uniform books per shelf equals HCF$(780, 364)$. We know:

Formula Used: Books per shelf $=$ HCF$(780, 364)$, Total shelves $= \dfrac{780 + 364}{\text{HCF}}$

Prime factorising 780 and 364:

$$780 = 2^2 \times 3 \times 5 \times 13, \qquad 364 = 2^2 \times 7 \times 13. \qquad \textbf{[1M]}$$

Common primes are 2 and 13, with smallest powers 2^2 and 13^1:

$$\text{HCF}(780, 364) = 2^2 \times 13 = 4 \times 13 = \boxed{52 \text{ books per shelf}}. \qquad \textbf{[1M]}$$

Total shelves $= \dfrac{780}{52} + \dfrac{364}{52} = 15 + 7 = \boxed{22 \text{ shelves}}$. **[1M]**

Q10. Prove that if p and q are distinct prime numbers, then $\sqrt{p} + \sqrt{q}$ is an irrational number. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Suppose, on the contrary, that $r = \sqrt{p} + \sqrt{q}$ is rational ($r \in \mathbb{Q},\ r > 0$). Writing $\sqrt{q} = r - \sqrt{p}$ and squaring both sides:

$$(\sqrt{q})^2 = (r - \sqrt{p})^2 \implies q = r^2 + p - 2r\sqrt{p}. \qquad \textbf{[1M]}$$

Isolating $\sqrt{p}$:

$$2r\sqrt{p} = r^2 + p - q \implies \sqrt{p} = \frac{r^2 + p - q}{2r}.$$

We know rational numbers are closed under addition, subtraction, multiplication, and non-zero division, while prime square roots are irrational:

Formula Used: $r \in \mathbb{Q} \setminus \{0\} \implies \dfrac{r^2 + p - q}{2r} \in \mathbb{Q}$, but $\sqrt{p} \notin \mathbb{Q}$ for prime p

Thus $\sqrt{p}$ must be rational. **[1M]**

Yet if $\sqrt{p} = \frac{a}{b}$ with HCF$(a, b) = 1$, then $a^2 = pb^2 \implies p \mid a$ and $p \mid b$, contradicting HCF$(a, b) = 1$.

Therefore $\boxed{\sqrt{p} + \sqrt{q} \text{ is irrational.}}$ **[1M]**

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Q11. A dairy distributes milk from three large containers holding 403 litres, 434 litres, and 465 litres of milk respectively. (i) Find the maximum capacity of a measuring container that can measure the milk of all three containers in an exact number of times. (ii) How many times must this measuring container be used to empty each container? (iii) Find the total number of times the measuring container is filled. [5]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

We know exact measurement without remainder requires a common divisor of $403, 434$, and 465:

Formula Used: Max measuring capacity $=$ HCF$(403, 434, 465)$

(i) Max capacity: Prime factorising $403, 434$, and 465:

$$403 = 13 \times 31, \qquad 434 = 2 \times 7 \times 31, \qquad 465 = 3 \times 5 \times 31. \qquad [1M]$$

Since 31 is the only common prime factor: HCF$(403, 434, 465) = \boxed{31 \text{ litres}}$. [1M]

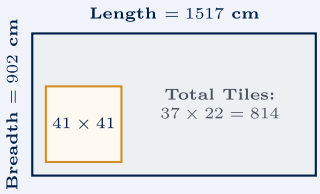
(ii) Number of measures per container: Dividing each volume by 31 L:

$$\text{Container 1: } \frac{403}{31} = \boxed{13 \text{ times}}, \qquad \text{Container 2: } \frac{434}{31} = \boxed{14 \text{ times}},$$
$$\text{Container 3: } \frac{465}{31} = \boxed{15 \text{ times}}. \qquad [2M]$$

(iii) Total fillings: $13 + 14 + 15 = \boxed{42 \text{ times}}$. [1M]

Q12. A warehouse floor of length 15 m 17 cm and breadth 9 m 2 cm is to be paved with the minimum number of identical square tiles. (i) Find the side length of the largest square tile that can be used. (ii) Calculate the exact number of tiles required. (iii) If each tile costs Rs 45, compute the total expenditure for paving the floor. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]



Unit Conversion: Length $L = 15$ m 17 cm $= 1517$ cm; Breadth $B = 9$ m 2 cm $= 902$ cm. Maximizing tile side s minimizes tile count.

We know the square-tiling HCF and area division formulas:

Formula Used: $s = \text{HCF}(L, B)$, $\quad \text{Tiles} = \dfrac{L \times B}{s^2}$, $\quad \text{Cost} = \text{Tiles} \times \text{Rate}$

(i) Largest tile side: Factorising $1517 = 37 \times 41$ and $902 = 2 \times 11 \times 41$:

$$s = \text{HCF}(1517, 902) = \boxed{41 \text{ cm}}. \qquad [2M]$$

(ii) Exact number of tiles:

$$\text{Tiles} = \left(\frac{1517}{41}\right) \times \left(\frac{902}{41}\right) = 37 \times 22 = \boxed{814 \text{ tiles}}. \qquad [1.5M]$$

(iii) Total expenditure at Rs 45 per tile:

$$\text{Total cost} = 814 \times 45 = 814 \times (40 + 5) = 32560 + 4070 = \boxed{\text{Rs } 36630}. \qquad [1.5M]$$

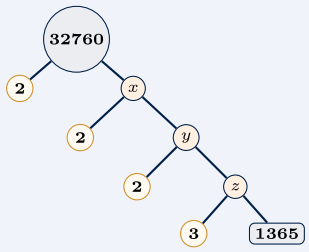
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Q13. CASE STUDY 09: MATHEMATICAL FACTOR TREE PRIME DECOMPOSITION [4 Marks]

During a mathematics laboratory session on the Fundamental Theorem of Arithmetic, a student investigates the prime factorisation of large composite integers by constructing recursive factor trees. For the number 32760, the student creates a tree where the number branches into 2 and a composite node x; node x branches into 2 and node y; node y branches into 2 and node z; and node z branches into 3 and a node $w = 1365$.

(i) Determine the numerical values of the missing nodes x, y, and z. [2]
(ii) Write the complete prime factorisation of 32760 in canonical exponential form. [1]
(iii) State the Fundamental Theorem of Arithmetic. [1]

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]



Factor Tree Structure: Starting at root 32760, successive divisions by $2, 2, 2$, and 3 trace
$x = 16380 \to y = 8190 \to z = 4095 \to w = 1365$.

We know each parent node equals the product of its child nodes:

Formula Used: Parent Node = Left Branch × Right Branch $\implies$ Missing Node = $\dfrac{\text{Parent}}{\text{Given Branch}}$

(i) Missing nodes x, y, z: Working down from 32760 (and verifying upward from $w = 1365$):

$$x = \frac{32760}{2} = \boxed{16380}, \qquad y = \frac{16380}{2} = \boxed{8190}, \qquad z = \frac{8190}{2} = \boxed{4095}. \qquad [2M]$$

Check: $3 \times w = 3 \times 1365 = 4095 = z$.

(ii) Canonical prime factorisation of 32760: Decomposing $1365 = 3 \times 455 = 3 \times 5 \times 91 = 3 \times 5 \times 7 \times 13$:

$$32760 = 2 \times 2 \times 2 \times 3 \times 3 \times 5 \times 7 \times 13 = \boxed{2^3 \times 3^2 \times 5 \times 7 \times 13}. \qquad [1M]$$

(iii) Fundamental Theorem of Arithmetic: Every composite integer can be expressed (factorised) as a product of primes, and this factorisation is $\boxed{\text{unique}}$, apart from the order in which the prime factors occur. [1M]

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